

Lab 1: Complete Lab-Focused Teaching Package

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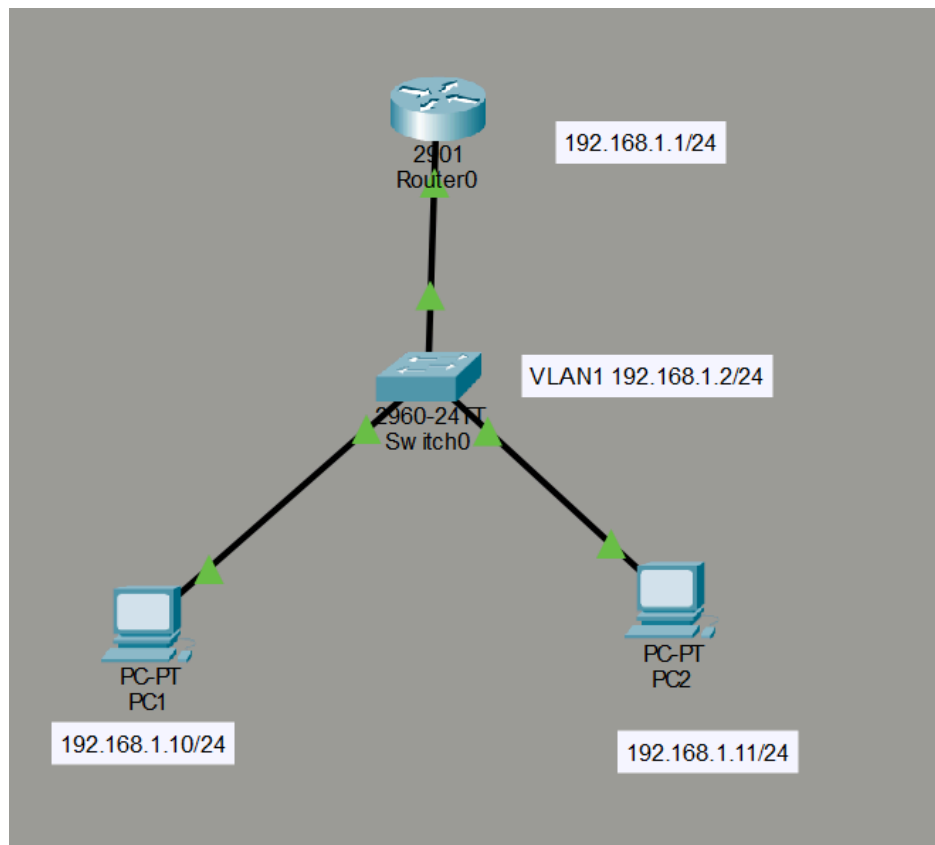
หน้าที่รับผิดชอบ

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PART 1: DEVICE CONFIGURATION

CISCO file

LAB1.pkt



ตารางข้อมูลที่ใช้ตั้งค่าเครือข่าย

Device	Interface	IP Address	Subnet Mask	Default Gateway	MAC (Dynamic)
R1	G0/0	192.168.1.1	255.255.255.0	N/A	0001.9643.AC01
S1	VLAN 1	192.168.1.2	255.255.255.0	192.168.1.1	0002.1745.8B02
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1	0050.7966.6803
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1	0090.2B14.7C04

Network: 192.168.1.0/24

Usable Host Range: 192.168.1.1 - 192.168.1.254

Broadcast: 192.168.1.255

PART 2: CONNECTIVITY VERIFICATION

PC2 ping PC1, Switch, Router

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=10ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC1 ping PC2, Switch, Router

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=13ms TTL=128
Reply from 192.168.1.10: bytes=32 time=5ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 5ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

ARP Table

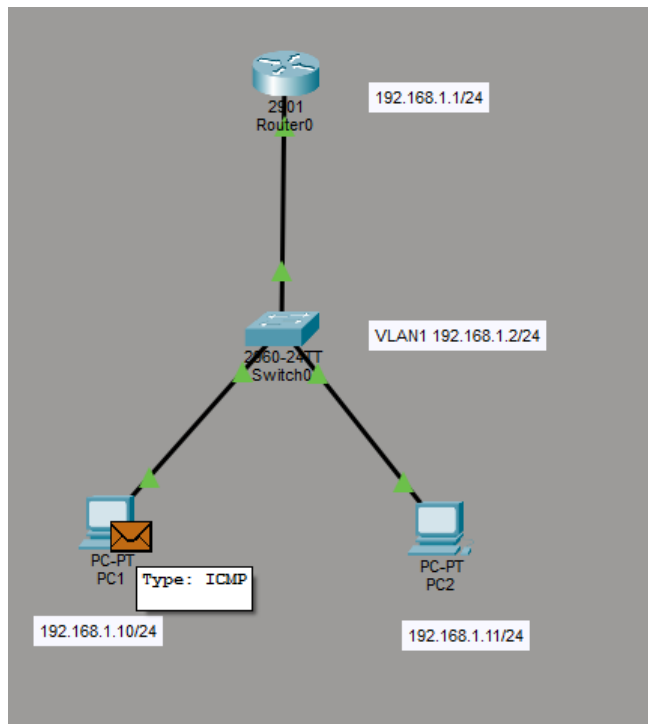
ARP Table for PC1		
IP Address	Hardware Address	Interface
192.168.1.1	0001.C973.2A01	FastEthernet0
192.168.1.2	0001.9658.7811	FastEthernet0
192.168.1.11	0001.64C7.5D93	FastEthernet0

ARP Table for PC2		
IP Address	Hardware Address	Interface
192.168.1.1	0001.C973.2A01	FastEthernet0
192.168.1.2	0001.9658.7811	FastEthernet0
192.168.1.10	0000.0C6A.9165	FastEthernet0

ARP Table for Switch0		
IP Address	Hardware Address	Interface
192.168.1.2	0001.9658.7811	Vlan1
192.168.1.10	0000.0C6A.9165	Vlan1
192.168.1.11	0001.64C7.5D93	Vlan1

ARP Table for Router0		
IP Address	Hardware Address	Interface
192.168.1.1	0001.C973.2A01	GigabitEthernet0/0
192.168.1.10	0000.0C6A.9165	GigabitEthernet0/0
192.168.1.11	0001.64C7.5D93	GigabitEthernet0/0

PART 3: ENCAPSULATION ANALYSIS



PDU Information at Device: PC1



OSI Model

Outbound PDU Details

Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer3	Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 192.168.1.11 ICMP Message Type: 8
Layer2	Layer 2: Ethernet II Header 0000.0C6A.9165 >> 0001.64C7.5D93
Layer1	Layer 1: Port(s): FastEthernet0

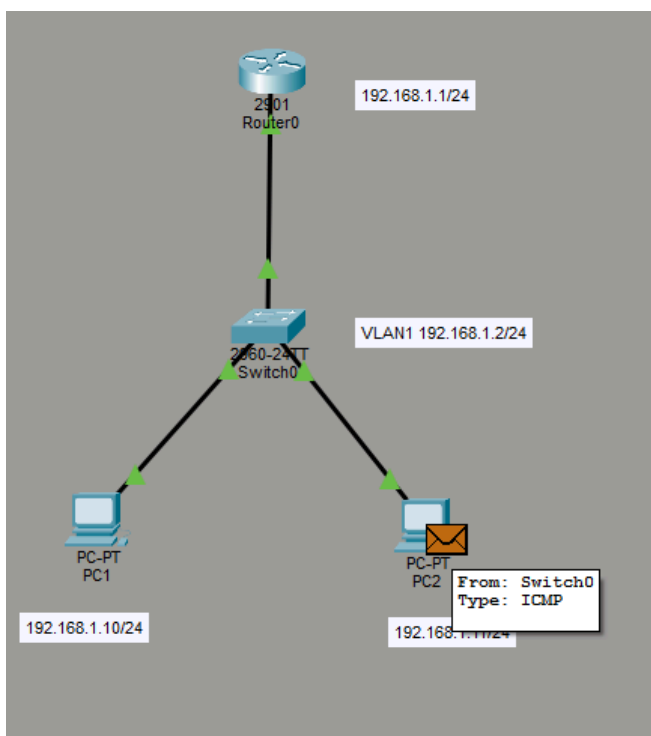
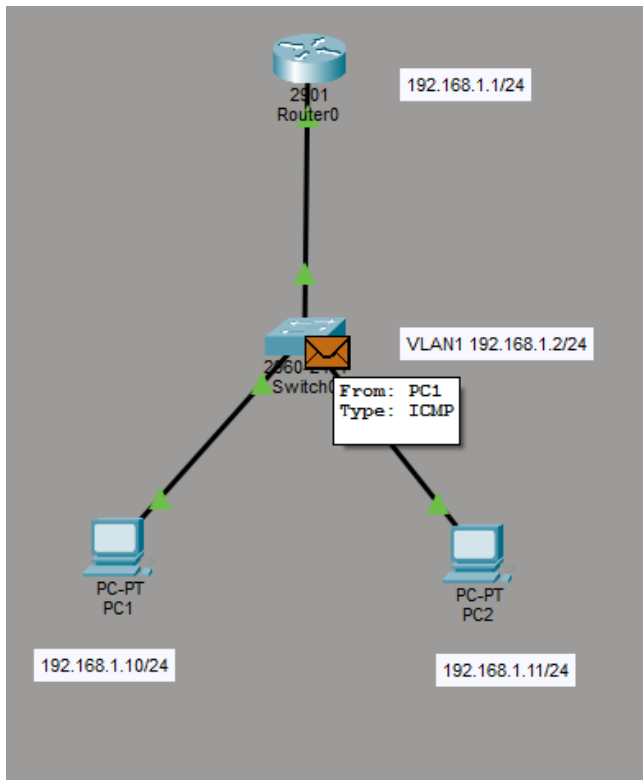


1. FastEthernet0 sends out the frame.

Challenge Me

<< Previous Layer

Next Layer >>

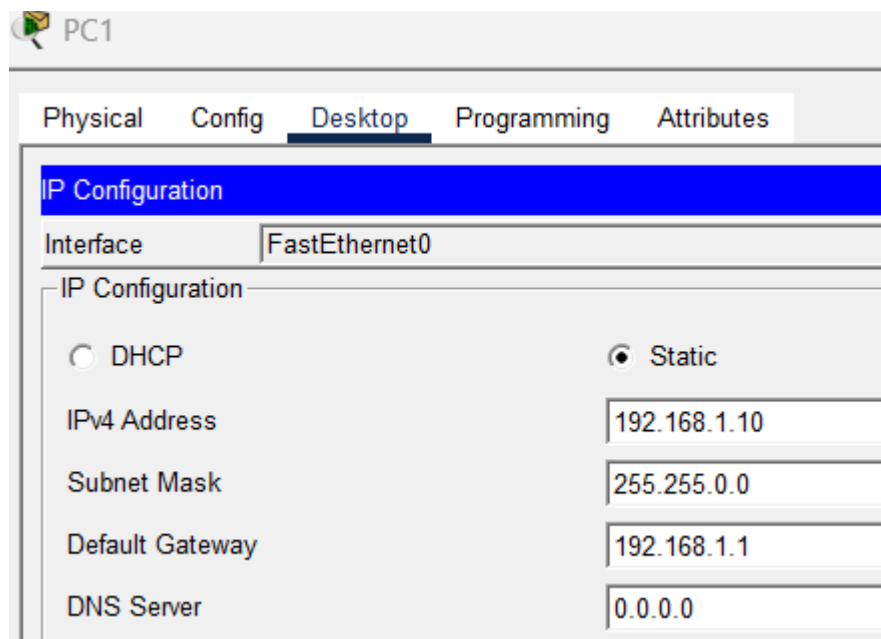


Step	Protocol	OSI Layer	TCP/IP Layer	Key Field Observed
1	ARP Request	Layer 2	Network Access	Target IP: 192.168.1.11
2	ARP Reply	Layer 2	Network Access	Sender MAC: 0001.64C7.5D93
3	ICMP Echo	Layer 3	Internet	Type: 8, Code: 0
4	Ethernet	Layer 2	Network Access	FCS: 0x00000000
5	ICMP Reply	Layer 3	Internet	Type: 0, Code: 0

PART 4: TROUBLESHOOTING CHALLENGE

Scenario A: Incorrect Subnet Mask

เปลี่ยน Subnet mark เป็น 255.255.0.0



ทดสอบ ping pc01 กับ pc2 แล้วสามารถ ping ได้

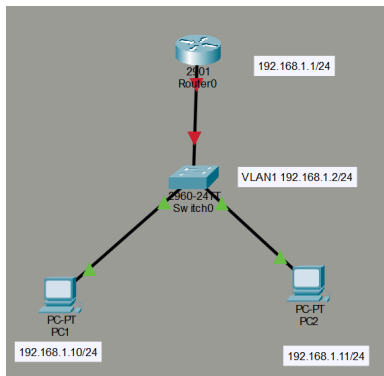
```
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

ใช้ arp -a แล้วขึ้น

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.1.11          0001.64c7.5d93        dynamic
```


Scenario B: Router Interface Shutdown



ไม่สามารถ ping Router ได้ แต่ยังสามารถ ping device อื่นได้

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

tracert Router

```
C:\>tracert 192.168.1.1

Tracing route to 192.168.1.1 over a maximum of 30 hops:

  1  *             *             *             Request timed out.
  2  *             *             *             Request timed out.
  3  *             *             *             Request timed out.
  4  *             *             *             Request timed out.
  5  *             *             *             Request timed out.
  6  *             *             *             Request timed out.
  7  *             *             *             Request timed out.
  8  *             *             *             Request timed out.
  9  *             *             *             Request timed out.
 10 *             *             *             Request timed out.
 11 *             *             *             Request timed out.
 12 *             *             *             Request timed out.
 13 *             *             *             Request timed out.
```

```
R1#show ip int brief

Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.1.1    YES NVRAM    administratively down down
GigabitEthernet0/1       unassigned      YES NVRAM    administratively down down
Vlan1                    unassigned      YES unset   administratively down down
R1#
```

Scenario C: Wrong Default Gateway

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: FE80::201:64FF:FEC7:5D93
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.11
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.1.99

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0
```

ยังสามารถ ping ภายใน LAN ได้

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time=9ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>
```

PC2 ping Router

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
C:\>tracert 192.168.1.2

Tracing route to 192.168.1.2 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.1.2

Trace complete.
```

PART 5: ADVANCED OBSERVATION (15 minutes)

ใส่ -l แล้วขึ้น invalid command เลยตัดทิ้ง

```
C:\>ping 192.168.1.11 -n 5

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Observe ARP

PC1 -> switch -> PC2 -> switch -> PC1

```
S1#clear mac-address-table dynamic
S1#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -

```

หลังจาก generate traffic

```
-----
S1#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0000.0c6a.9165    DYNAMIC Fa0/1
      1    0001.64c7.5d93    DYNAMIC Fa0/2
      1    0001.c973.2a01    DYNAMIC Gig0/1
S1#
```

Document?

Application data generation	ส่งค่าขอตรวจสอบการเชื่อมต่อระหว่างเครื่อง ด้วยคำสั่ง ping	ICMP Type 8 (echo)
TCP segment creation	-	-
IP Packet Encapsulation	เอาข้อมูลที่ได้จาก ค่าของมาห่อหุ้มด้วย IP header เพื่อที่จะสามารถระบุ IP ได้	IPv4
Ethernet framing	นำข้อมูลมาเพื่อหาปลายทาง	Ethernet II Header
Physical Transmission	สัญญาณไฟฟ้า?	Port(ไม่ระบุ)

LAB DELIVERABLES

Worksheet 1: Configuration Verification

Checkpoint	Command Used	Expected Result	Actual Result	Pass/Fail
R1 Interface Status	<code>show ip int brief</code>	G0/0 UP/UP	G0/0 UP/UP G0/1 DOWN/DOWN	pass
S1 VLAN Config	<code>show vlan brief</code>	VLAN 1 Active	VLAN 1 Active	pass
PC1 Connectivity	<code>ping 192.168.1.11</code>	Success (4 replies)	4 replies	pass
ARP Resolution	<code>arp -a</code> on PC1	2+ entries	3 entries	pass
MAC Learning	<code>show mac address-table</code> on S1	2 dynamic entries	3 entries	pass

Worksheet 2: Protocol Analysis

Capture this data during simulation:

1. ARP Request Frame:

- Source MAC: _____ 0000.0C6A.9165 _____
- Destination MAC: _____ 192.168.1.10 _____
- ARP Opcode: _____ 1 _____

2. ICMP Echo Packet:

- Time to Live (TTL): _____ 255 _____

- Protocol Field: _____ 1 _____ (Value = 1)

3. **Encapsulation Chain:** Data → ____TCP/UDP____ Header → ____IP____ Header → ____Ethernet____ Header → Bits

Worksheet 3: Troubleshooting Journal

Problem Injected	Symptoms Observed	Diagnostic Commands Used	Root Cause	Solution
Wrong Subnet Mask	ยังสามารถที่จะ ping ได้	<code>ping, arp -a</code>	Subnet Mask ของ PC1 ที่ เปลี่ยนเป็น 255.255.0.0 ซึ่ง มีขอบเขตกว้าง กว่าค่าที่ถูกต้อง ทำให้ยังมองว่าอยู่ในเครือข่าย เดียวกัน	Change back to 255.255.255.0
Interface Shutdown	can't ping R1 or PC1.	<code>show ip int brief</code>	Router DOWN	R1(config-if)# no shutdown
Incorrect Gateway	No noticeable symptoms.	<code>ipconfig, tracert</code>	Gateway doesn't affect LAN	Change back to 192.168.1.1

ASSESSMENT QUESTIONS

- When PC1 pings PC2, which protocol resolves the MAC address before ICMP operates?
Ans: ARP, because you need MAC address(Layer 2) before IP packet(Layer 3) which contain ICMP data.
- What would happen if you changed the subnet mask on PC1 to 255.255.255.128?
Ans: Can still ping in LAN, no noticeable symptoms.
- Why does the switch not need an IP address for basic frame forwarding?
Ans: It operate at layer 2, which use MAC address.
- Which OSI layers are involved when R1 forwards a packet between interfaces?
Ans: OSI layer 3.
- What is the purpose of the TTL field in the IP header you observed?

Ans: Prevent loop in network by decrementing until 0.
